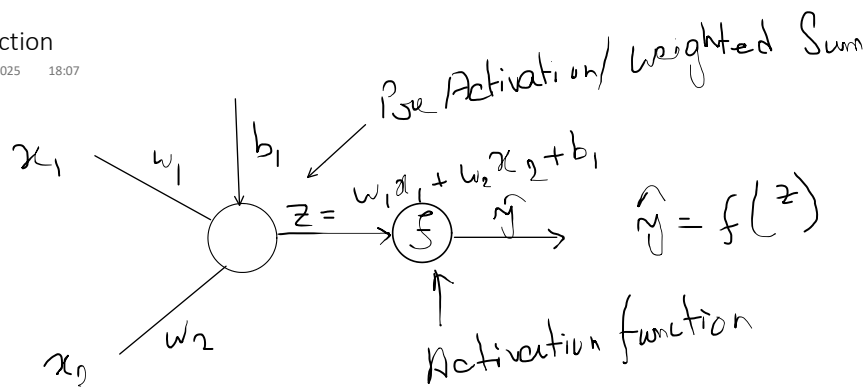
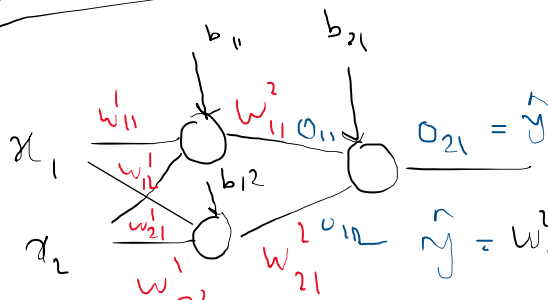




Why?

1. to capture non linearity without Act. Func.



$$\hat{y} = w_{11}^2 (w_{11}^1 x_1 + w_{21}^1 x_2 + b_{11}) + w_{21}^2 (w_{12}^1 x_1 + w_{22}^1 x_2 + b_{12}) + b_{21}$$

$$\hat{y} = w_{11}^2 w_{11}^1 x_1 + w_{11}^2 w_{21}^1 x_2 + w_{11}^2 b_{11} + w_{21}^2 w_{12}^1 x_1 + w_{21}^2 w_{22}^1 x_2 + w_{21}^2 b_{12} + b_{21}$$

Linear * without Activation function, non linearity cannot be captured *

Characteristics:

1. Non linear function

... linear ... → Desirable

... (0, 1)

1. Non linear → Desirable
2. Differentiable → Desirable
↳ Not mandatory (eg. ReLU)
3. Computationally inexpensive.
4. Zero centered → Desirable
↳ Not mandatory (eg. Sigmoid)
 $\text{mean}(\text{activation function of an layer}) = 0$
5. Non Saturating → Desirable
↳ Not mandatory (eg. tanh, Sigmoid)

Saturating for any large i/p
O/p will not be squished

$\left. \begin{array}{l} \text{tanh} \rightarrow [-1, +1] \\ \text{sigmoid} \rightarrow (0, 1) \end{array} \right\} \text{Vanishing Gradient Problem.}$

④ Farrow Activation function

Sigmoid | $\sigma(z) = \frac{1}{1 + e^{-z}}$

$$z = e^{-z}$$

Derivative

$$\sigma'(z) = \sigma(z) [1 - \sigma(z)]$$

$$\tanh'(z) = 1 - \tanh^2(z)$$

$$\tanh \mid \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

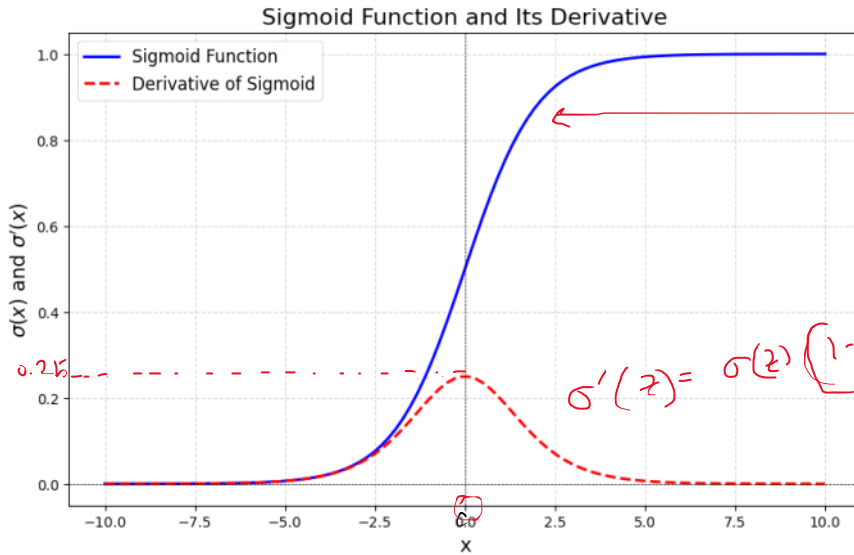
$$\text{relu} \mid \text{Relu} = \max(0, z)$$

$$\tanh'(z) = 1 - \tanh^2(z)$$

$$\text{relu}'(z) = \begin{cases} 0 & z \leq 0 \\ 1 & z > 0 \end{cases}$$

Sigmoid Activation Function

22 December 2025 20:23



Advantages:-

1. $(0, 1)$ ← treat as Probability

O/P layer ← Binary Classification

2. Non Linear

3. Differentiable

Disadvantages:-

1. Saturating

function:-

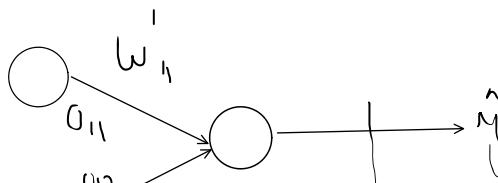
$$\sigma(-\infty, +\infty) = (0, 1)$$

* Not used in Hidden layers.

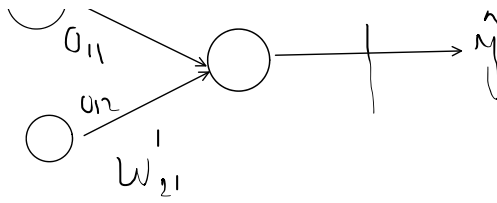
Vanishing Gradient

2. Non zero centered

$$x + b_{11}$$



$$z = w'_{11}O_{11} + w'_{21}O_{12} + b_{11}$$



$$\frac{dz}{dw'_{11}} = O_{11}$$

$$\frac{dz}{dw'_{21}} = O_{12}$$

$$O_{11} = \sigma(\text{Some})$$

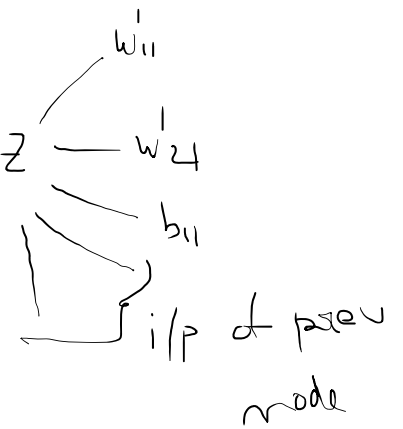
$$O_{12} = \sigma(\text{Some})$$

+/- ✓

$$\frac{dL}{dw'_{11}}$$

$$= \frac{dL}{dy} \cdot \frac{dy}{da} \cdot \frac{da}{dz}$$

$$\left| \frac{dz}{dw'_{11}} \right|$$



$$\frac{dL}{dw'_{21}}$$

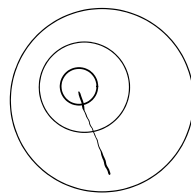
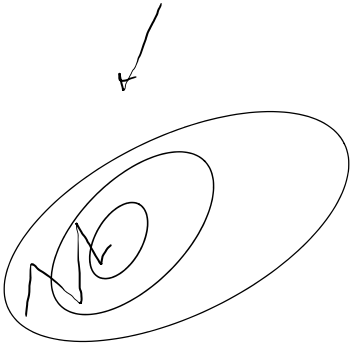
$$= \frac{dL}{dy} \cdot \frac{dy}{da} \cdot \frac{da}{dz}$$

$$\frac{dz}{dw'_{21}}$$

↑
+ve as it's $\sigma'(\text{Some})$

$W \rightarrow \begin{matrix} \uparrow \\ \downarrow \end{matrix} \left\{ \text{At once} \right\}$

$W \rightarrow \begin{matrix} \uparrow \\ \downarrow \end{matrix} \left\{ \text{Simultaneously} \right\}$



$$w'_i = w_i - \eta \frac{\partial L}{\partial w_i} \Rightarrow \underline{w'_i - w_i} = -\eta \frac{\partial L}{\partial w_i}$$

$$\Rightarrow \boxed{\Delta w_i = -\eta \frac{\partial L}{\partial w_i}}$$

$$\frac{\partial L}{\partial w_i} = \frac{\partial L}{dy} \cdot \frac{dy}{dz} \cdot \left(\frac{dz}{dw_i} \right)$$

z = i/p or activation from prev layer.

$$\frac{\partial \mathcal{L}}{\partial w_i} = \frac{\partial \hat{y}}{\partial z} \left(\frac{\partial \mathcal{L}}{\partial z} \right)$$

Backpropagation Error $\rightarrow \delta$

prev layer $\rightarrow x_i$

$$\Delta w_i^o = -\eta \delta x_i^o$$

for non zero centered activation,

$$x_i^o > 0$$

$$\delta > 0 \rightarrow \Delta w_i^o = (-)$$

δ $\rightarrow$ > 0 ; $\Delta w_i^o = (-)$

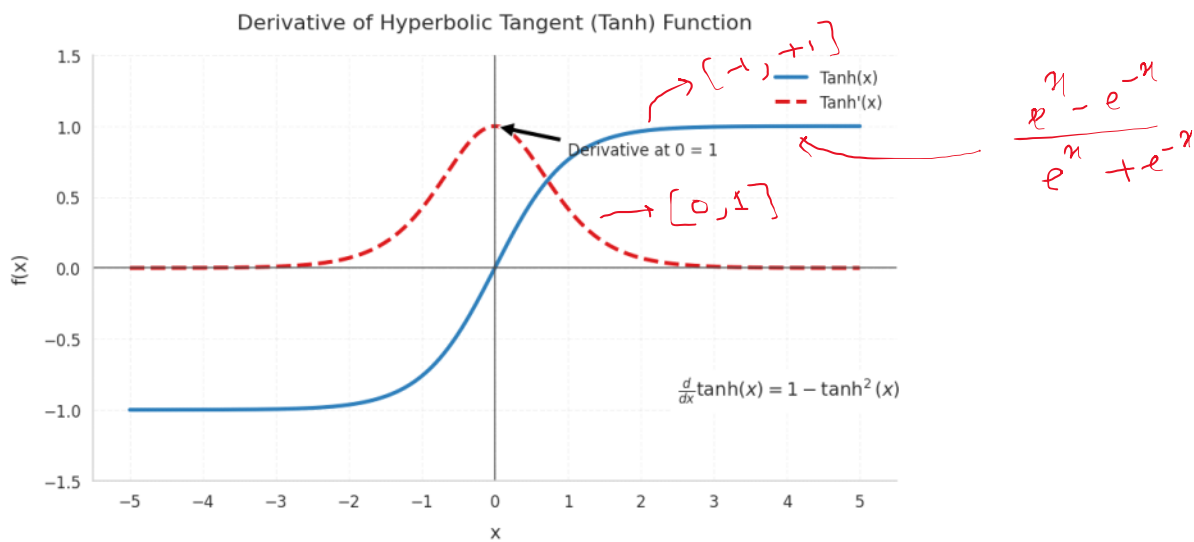
$$\delta < 0 \rightarrow \Delta w_i^o = (+ve)$$

$\Delta w_i^o \rightarrow +ve$
 $\Delta w_i^o \rightarrow -ve$ } for same layer for all weights

3. Computationally expensive

Tanh Activation Function

22 December 2025 20:58



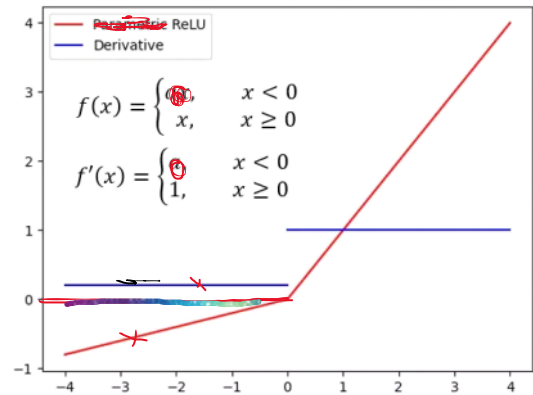
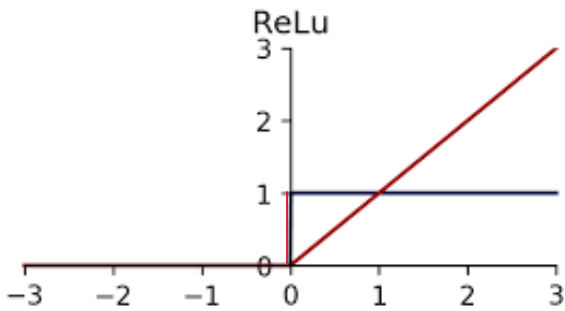
Advantage:-

1. Not linear
2. Differentiable
3. Zero centered (Smooth training)

Disadvantage:-

1. Saturating function (Vanishing Gradient Descent)
2. Computationally expensive

Rectified Linear Unit



$$f(x) = \max(0, x)$$

Advantages:-

1. Non Linear

$$f(x) = \max(0, x+1) - \max(0, x-1)$$

$f(x)$ is Linear

1. Additive Property

2. Homogeneity

$$f(x_1 + x_2) = f(x_1) + f(x_2)$$

$$f(ax) = af(x)$$

for relu $x_1 = -1$ $x_2 = 2$

(1) $\text{Relu}(x_1 + x_2) = \text{Relu}(-1 + 2) = \text{Relu}(1) = 1$

~~(2)~~ $\text{Relu}(x_1) + \text{Relu}(x_2) = \text{Relu}(-1) + \text{Relu}(2) = 0 + 2 = 2$

(1) $x = -2$ $a = 3$

$\text{Relu}(ax) = \text{Relu}(3 \cdot -2) = \text{Relu}(-6) = 0$

$\text{Relu}(x) = \text{Relu}(-2) = 0$

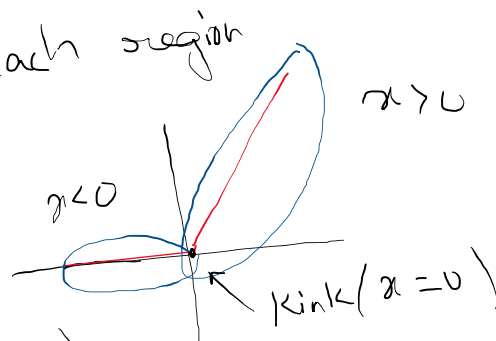
$$\alpha \text{Relu}(x) = \text{Relu}(-x) = 0$$

$$x=1 \quad \alpha=-1$$

$$\text{Relu}(\alpha x) = 0 \neq$$

$$\alpha \text{Relu}(x) = -1$$

* Relu is linear in each region
not globally



$$f(x) = \max(0, x+1) - \max(0, x-1)$$

$$= 0 \text{ when } x \leq -1$$

$$= x+1 \text{ when } x > -1$$

$$= 0 \text{ when } x \leq 1$$

$$= x-1 \text{ when } x > 1$$

Case I

$$x \leq -1$$

$$\max(0, x+1) = 0$$

$$\max(0, x-1) = 0$$

$$f(x) = 0$$

Case II

$$-1 < x \leq 1$$

$$\max(0, x+1) = x+1$$

$$\max(0, x-1) = 0$$

$$f(x) = x+1$$

Case III

$$x > 1$$

$$\max(0, x+1) = x+1$$

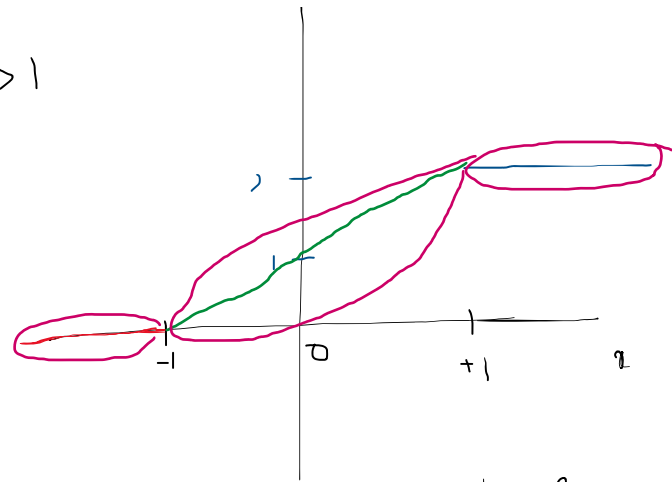
$$\max(0, x-1) = x-1$$

$$f(x) = 2$$

$$f(x) = \begin{cases} 0 & x \leq -1 \\ x+1 & -1 < x \leq 1 \\ x-1 & x > 1 \end{cases}$$

$$h(x) = 2f(x+1) - 5f(x-1)$$

$x > 1$



Advantage:-

1. Non linear
2. Non Saturated in the +ve Region
3. Computationally inexpensive
4. Fast Convergence

Disadvantage

1. Not differentiable at $x=0$
2. Not ^{Batch Normalization} zero centered

3. Dying ReLU

Use
diff Variant
of ReLU